INCH-POUND

MIL-C-71090 (AR)
AMENDMENT 2
29 September 1994
SUPERSEDING
AMENDMENT 1
28 April 1992

MILITARY SPECIFICATION

CANISTER SMOKE, ASSEMBLY
FOR PROJECTILE, PRACTICE, 155MM, M804E1

This amendment forms a part of Military Specification MIL-C-71090 (AR), dated 15 November 1991, and is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Title: Change, "M804E1" to "M804A1"

PAGE 2

2.1.1 STANDARDS, MILITARY

Delete:

"MIL-STD-1218 - ACS Chemicals"

2.2 Non-government publications, ASTM

Delete:

"ASTM D2905 - Statement on Number of Specimens Required to Determine the Average Quality of Textile"

PAGE 3

3.3 Delete in its entirety and substitute the following:

"3.3 Smoke canister seal. The smoke canister assembly shall withstand a vacuum of 6.0 +/- 1.0 inches of mercury for a minimum period of 30 seconds without leakage."

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3.7 Add the following:

"3.7 Proving ground. The smoke canister, when assembled into a projectile, fuzed, and tested as specified, shall upon ground impact functioning produce a smoke signature visible across 4000 meters of level terrain."

3.8 Add the following:

"3.8 Smoke mix process control. The process for weighing, mixing and dispensing the smoke mix and constituents thereof shall be established, and qualified by the first article proving ground test. Once qualified, changes to the equipment or procedures shall require prior approval of the contracting officer. Re-qualification shall also be required if in the opinion of the Government procuring activity, maintenance or repair to the mixing or dispensing mechanism effects process control parameters. The following process controls are mandatory:

a. Weigh and check weigh each constituent prior to addition to blender. The check weigh shall be a different device by a different operator or inspector.

b. Minimum mixing time shall be that used to mix the first article (qualification) batch(s).

c. Mixing speed shall be within plus or minus 10% of that used to mix the first article (qualification) batch(s)."

PAGE 4

4.3.1, Line 6 Under "Quantity" delete "20" and substitute "25"

4.3.2.1, Line 2, Delete "composition and"

PAGE 5

Delete "TABLE I" and substitute the following included "TABLE I."

PAGE 11

* 4.4.2.4 Delete Major Defects 101 and 102 in their entirety and substitute:

"101	Smoke canister seal	Note 3,	3.3,	4.5.3
102	Mix process incorrect	Note 1,	3.8,	Visual"

Add Note 3:

"3. Sampling in accordance with 4.4.3.3."

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Canister Smoke, Assembly and Components	1 OF 1	See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Canister Smoke Assembly (Dwg. 12913950) Examination for defects (Except for Proving Ground Test) Proving Ground Test	 5, 0-1 20, 0-1	 3.2 3.7	 4.4.2.4 4.5.1
	Components Pad (Dwg. 9349934) Examination for defects	 5, 0-1	 3.2	 4.4.2.1
	Cap (Dwg. 9349935) Examination for defects	 5, 0-1	 3.2	 4.4.2.2
	Canister (Dwg. 9349936) Examination for defects	 5, 0-1	 3.2	 4.4.2.3

NOTES:

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Delete 4.4.3.1, 4.4.3.1.1, 4.4.3.1.2, 4.4.3.1.2.1, 4.4.3.1.2.2 in their entirety and substitute:

"4.4.3.1 Mixing process control. Prior to dispensing each batch (see 6.8) of smoke mix, the following process control checks shall be made:

a. Check records of weigh and check weigh of all constituents.

b. Check mixing time, minimum.

c. Check mixing speed record for entire minimum time.

d. Examine weighing and mixing area for evidence of spilled constituents in excess of 1% of batch weight. (Do not include dispensing area.)

If mixing time or speed are not as required, re-mix the batch for the required minimum time at the required speed. If the weigh and check weigh was not accomplished for any constituent, or if spilled constituents are in evidence, the batch shall be rejected."

PAGE 14

Delete 4.4.3.1.2.3 in its entirety.

4.4.3.3 Add the following:

"4.4.3.3 Smoke canister seal. A sample of 8 smoke canister assemblies shall be taken, for leakage testing, produced from each seaming machine, from each eight (8) hour shift or fraction of a shift from machines in use. The randomly selected smoke canister assemblies to be used as samples and all smoke canister assemblies they represent, shall be identified for each machine and shift until all tests and examinations of 4.4.3.3 are complete. Component parts selected for assembly into leak test samples shall be loaded with inert composition and marked as inert prior to seaming. The leak test samples shall not be returned to the inspection lot."

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Delete 4.5.1, 4.5.1.1 and 4.5.1.1.1 in their entirety and substitute:

* "4.5.1 Proving ground. The sample smoke canister assemblies shall be assembled into M804A1, 155MM, projectiles and tested at a Government proving ground in accordance with the applicable acceptance test procedure. Projectiles shall be fuzed with an M739 series PD fuze set "PD". The projectiles shall be fired in a M185 or M199 Series cannon using a M119A1, Zone 8 charge. Chamber pressures will be based on copper pressure gages. The projectiles and propelling charges shall be uniformly temperature conditioned to 21 degrees Celsius (70 degrees F) plus or minus 1 degree Celsius (2 degrees F) for a minimum of 24 hours prior to testing. Adequate photographic and instrumental coverage shall be provided to assure compliance with requirements. Range and deflections shall be recorded for informational purposes only. An observer shall be stationed 4000 ± 200 meters from the edge of the impact area to record presence of smoke signature. The test shall be conducted on level terrain and the wind velocity shall not exceed 15 knots at the time of testing."

PAGE 17

Delete 4.5.1.1.2 and 4.5.1.1.3 in their entirety.

PAGE 18

4.5.3 Delete in its entirety and substitute the following:

"4.5.3 Smoke canister seal. Place inert smoke canister assembly in fixture and close lid. Start vacuum. Lower canister into water. After gage shows 6.0 +/-1" of mercury, observe canister for a minimum of 30 seconds. Raise the canister out of the water. Release vacuum. Raise the lid, remove the canister, and dry off. Set aside the canister for scrap. If the canister is a reject, then identify the seaming machine that produced the reject, and reject all canisters produced by that seaming machine since the last acceptable leak test sample."

PAGE 19

Delete 6.2 d., 6.2 e. and 6.2 f. in their entirety.

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PAGE 20

6.6, Line 6:
Delete: "PMCAR-CCH-P"
Substitute: "SMCAR-FSA-MS"

6.10 Delete in its entirety.

PAGE 21

6.12 Delete: "Air Pressure test"
Substitute: "Smoke canister seal"

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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